Building Agentic Applications with LangChain and Deep Agents

1. Introduction

Purpose: Establish what students already know about LLMs and AI-powered applications, and get them thinking about how these application have evolved.

1.1. Discussion

- What are the chatbots you used?
- How long you have been using?
- Have you noticed how these applications changed over last few years?
- What can today's AI assistants do that earlier chatbots couldn't?

Earlier chatbots	Modern AI assistants
Primarily text-based conversations	Multimodal interactions
Respond to individual prompts	Handle multi-step tasks
Limited external capabilities	Can interact with external tools
Mostly stateless interactions	Can maintain conversational context
Generate responses	Can take actions on behalf of users

Can AI perform actions for us instead of just answering?

2. What is an Agent?

Purpose: Introduce to agentic systems.

1. What is an agent?
   Agent is a AI systems that can reason about a task, decide what actions to take, interact with its environment, and work towards achieving a specified goal.
2. What makes an application agentic?
   Application becomes agentic when it gives the AI system the ability to deterimine and execute actions towards goal.
   Examples:
   - Reading emails and identifying those requiring attention.
   - Writing and executing code, then examining the execution output.
   - Searching multiple sources and synthesizing information.
   - Interacting with external services through tools and APIs.

3. What is LangChain

Purpose: Introduce to LangChain

LangChain is an open-source framework and ecosystem for developing applications powered by Large Language Models (LLMs). It provides abstractions and integrations that simplify working with models, tools, data sources and agentic workflows.

Component	What it provides
LangChain	High-level abstractions for building LLM applications and agents
LangGraph	Graph-based orchestration for stateful and long-running agent workflows
Deep Agents	A higher-level agent harness built on LangGraph
LangSmith	Observability, evaluation and deployment capabilities